

Part A: Literature Exploration and Comparison

GROUP_163 DNN Assignment

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For our literature exploration and comparative study on the application of deep learning networks in specialized domains, we selected three representative papers each targeting a distinct area and architecture.

We chose the first paper for its innovative use of customized LSTM and CNN models to address the complex task of short-term electricity consumption forecasting, demonstrating how ensemble learning enhances predictive accuracy.

The second paper was selected because it applies state-of-the-art Vision and Recurrent Transformer architectures for human activity recognition in videos, showcasing the transformative impact of transformer models in sequential image analysis.

Our third choice is a comprehensive review that examines a wide range of deep learning and classical techniques for medical image identification, offering a

holistic perspective on current technologies and future challenges in medical imaging.

Selected Papers and Links

- **PAPER1:** Short-term electricity consumption forecasting with deep learning - CNN
<https://www.bohrium.com/paper-details/short-term-electricity-consumption-forecasting-with-deep-learning/1147845977722322947-2679>
- **PAPER2:** ViT-ReT: Vision and Recurrent Transformer Neural Networks for Human Activity Recognition in Videos
<https://arxiv.org/pdf/2208.07929.pdf>
- **PAPER3:** Medical image identification methods: A review
<https://pubmed.ncbi.nlm.nih.gov/38104516/>

Comparison table

Content name	Paper-1	Paper-2	Paper-3
Title of the paper	Short-term electricity consumption forecasting with deep learning - CNN	ViT-ReT: Vision and Recurrent Transformer Neural Networks for Human Activity Recognition in Videos	Medical image identification methods: A review

Authors	Emrah Demir, Serkan Gunal	James Wensel, Hayat Ullah, Arslan Munir	Juan Li, Pan Jiang, Qing An, Gai-Ge Wang, Hua-Feng Kong
Year of publication	2025	2022	2023
Network used	Customized LSTM, Customized CNN, and an Ensemble of both	Vision Transformer (ViT), Recurrent Transformer (ReT) neural networks. The system replaces traditional CNN and RNN chains in activity recognition models with specialized transformer architectures.	Survey and taxonomy of deep learning architectures including CNNs, RNNs, transfer learning models, hybrid networks, and classical machine learning for medical image identification and analysis.
Depth of the network	LSTM: 1 layer with 14 units CNN: 4 Conv2D layers with increasing filters (4, 8, 16, 32)	ViT: Patch embedding, multi-headed attention, feed-forward layers. ReT: Stack of transformer encoder modules	Various depths depending on the technique; reported models range from shallow classifiers to very deep CNN architectures (e.g., ResNet, VGG, U-Net) for image

		for sequence modeling; absence of decoder for classification.	classification, segmentation and detection.
How is the network helping the overall task? eg: feature engg or classification or regression or all	The networks are designed for short-term (24-hour) electricity consumption forecasting, which is a regression task. The customized LSTM employs the Rectified Linear Unit (ReLU) activation function throughout its gating mechanisms instead of the traditional hyperbolic tangent (tanh) function. This modification mitigates vanishing gradient issues and accelerates convergence, resulting in improved training	The ViT extracts spatial features from image frames by treating images as sequences of patches and employing attention mechanisms for robust feature embeddings. ReT specializes in capturing temporal dependencies in sequences of video frames. The combination allows for efficient and scalable human activity recognition, leveraging transformers to replace computationally costly CNN/RNN stages while maintaining high	Deep learning and hybrid methods enable feature extraction (CNNs), temporal modeling (RNNs), and classification tasks for medical image identification, segmentation, detection, and registration. Transfer learning and hybrid networks improve generalizability and robustness, especially with limited annotated medical data. Task categories include object detection, classification, segmentation, and image registration.

	stability for electricity consumption forecasting. The customized CNN utilizes Conv2D and MaxPooling2D layers with input data structured as (168, 7, 1) to represent 168 hours of data across 7 features. This approach is critical for preserving multivariate feature relationships, preventing information loss across multiple attributes that can occur when using Conv1D layers. This architectural refinement significantly enhanced the prediction performance of the	classification accuracy.	
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	CNN model when dealing with multiple input features. The ensemble model combines the individual predictions of the customized LSTM and CNN models using a weighted averaging approach. This method enhances accuracy, generalizability, and robustness by leveraging the individual strengths of both constituent models.		
Loss function used	Mean Squared Error (MSE)	Cross-Entropy loss for multi-class activity recognition (classification).	Various: Cross-Entropy for classification, Dice coefficient for segmentation, Mean Squared Error, and other domain-specific metrics.

Evaluation / Performance metric used	Normalized RMSE (N-RMSE), Normalized MAE (N-MAE), Mean Absolute Percentage Error (MAPE)	Accuracy, Top-1/Top-5 accuracy, precision, recall, F1-score; extensive benchmarks on UCF101 dataset for activity recognition.	Accuracy, sensitivity, specificity, precision, recall, F1-score, area under curve (AUC) depending on the application (classification, segmentation, detection, etc.).
Name of Dataset used. If a public dataset, provide the URL.	Hourly electricity consumption data for Dayton, Ohio, sourced from Kaggle and PJM Interconnection. Meteorological variables (temperature, relative humidity, and wind speed) for Dayton were obtained from Open-Meteo. Hourly electricity consumption data for Houston, Texas, sourced from Kaggle.	URL: https://www.crcv.ucf.edu/data/UCF101.php	Dataset varies by application: Classification (ImageNet, CheXpert, ISIC, BraTS), segmentation (BraTS, LiTS, KiTS), detection (DRIVE, LUNA16). Most datasets are publicly available from medical repositories.

	Meteorological parameters for Houston were also collected from Open-Meteo		
Conclusion	Ensemble models combining LSTM and CNN outperform standalone models. CNN achieved best accuracy (MAPE as low as 4.06%). Meteorological and timestamp features significantly improve forecasting. The models generalize well across cities and seasons, offering practical value for energy management.	ViT-ReT hybrid transformer models offer scalable and accurate human activity recognition, outperforming traditional CNN/RNN chains while reducing computational complexity and providing faster inference suitable for edge devices and IoT systems.	Deep learning models, especially CNNs and transformer-based architectures, have revolutionized medical image identification. They provide robust tools for classification, segmentation, and detection, though challenges remain with data diversity, annotation scarcity, and fairness. Hybrid and transfer learning models improve accuracy and generalizability. Future directions emphasize multimodal data integration and fair, interpretable AI.

Justification to choose PAPER-1 for implementation:

We chose the paper “Short-term Electricity Consumption Forecasting with Deep Learning” because it focuses on solving a real-world problem that is highly relevant to industries and power grid management.

Accurate short-term forecasting of electricity demand is crucial for operational efficiency, planning, and reducing costs.

The paper’s emphasis on deep learning, particularly RNN-based approaches, aligns with the objectives of our Deep Neural Networks course, giving us a practical way to apply and understand the concepts we are learning.

Another reason for selecting this paper was its clear methodology and potential for extension. While the original study primarily used RNNs, we saw an opportunity to expand it by testing CNN, LSTM, and Transformer models for comparison.

This not only deepened our technical understanding but also helped us generate new insights. The availability of a well-structured Kaggle dataset made implementation easier and reproducible, making this paper a perfect fit for our assignment goals.